1. Найти вычеты $f(z)$ отн. всех ее особых точек: $f(z) = \frac{1+\cos z}{z^2}$;

$z^2 = 0$; $z = 0$

$g(z) = \frac{1}{f(z)} = \frac{z^2}{1+\cos z}$, $\frac{1}{1+\cos z} \neq 0$ при $z = 0$

$z = 0$ – ноль 2-ого порядка для $g(z)$,

полюс 2-ого порядка для $f(z)$,

$$\operatorname*{Res}_{z=0} f(z) = \frac{1}{(2-1)!} \cdot \lim_{z\to 0} \frac{d}{dz}\left(z^2 \cdot \frac{1+\cos z}{z^2}\right) =$$

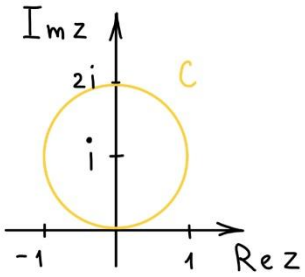
$$\lim_{z\to 0}(1+\cos z)' = \lim_{z\to 0}(-\sin z) = 0$$

2. Вычислить $\oint\limits_C \frac{(z^2-1)dz}{z^2+1}$; где $C = \{(x,y); y^2 + x^2 = 2y\}$

$C: y^2 + x^2 = 2y$

$C: y^2 - 2y + 1 + x^2 = 1$

$C: (y-1)^2 + x^2 = 1$



$C: |z - i| = 1$

$z^2 + 1 = 0$, $z = \sqrt{-1} = \pm i$

$z = -i$ не в контуре $\Rightarrow$ не рассматриваем

$z = i$ – полюс 1-ого порядка

$$\operatorname*{Res}_{z=i} \frac{(z^2-1)}{z^2+1} = \left.\frac{z^2-1}{2z}\right|_{z=i} = \frac{-2}{2i} = i$$

$$\oint\limits_C \frac{z^2-1}{z^2+1}\,dz = 2\pi i \cdot i = -2\pi$$

3. Вычислить $\int\limits_0^\infty \frac{(x^2+1)dx}{x^6+1}$

$$f(z) = \frac{z^2+1}{z^6+1}$$

$$z^6+1=0 \qquad x = \sqrt[6]{-1} = e^{\frac{i(\pi+2\pi k)}{6}} = e^{\frac{i\pi(1+2k)}{6}}$$

$$z_0 = e^{i\frac{\pi}{6}} \qquad z_1 = e^{i\frac{3\pi}{6}} = i \qquad z_2 = e^{i\frac{5\pi}{6}} \text{ – выше } \operatorname{Im} z = 0$$

$$z_3 = e^{i\frac{7\pi}{6}} \qquad z_4 = e^{i\frac{9\pi}{6}} = -i \qquad z_5 = e^{i\frac{11\pi}{6}} \text{ – ниже } \operatorname{Im} z = 0$$

$z_0 \ldots z_5$ – полюсы 1-ого порядка

$$\operatorname*{Res}_{z=z_k} f(z) = \frac{z^2+1}{6z^5}\Big|_{z=z_k}$$

$$\operatorname*{Res}_{z=z_0} f(z) = \frac{e^{i\frac{\pi}{3}}+1}{6e^{i5\frac{\pi}{6}}} = \frac{e^{-\frac{1}{2}\pi i}}{6} + \frac{1}{6e^{i5\frac{\pi}{6}}} =$$

$$\frac{1}{6}\left(-i+\cos\left(-\frac{5}{6}\pi\right)+i\sin\left(-\frac{5}{6}\pi\right)\right) = \frac{1}{6}\left(-\cos\left(\frac{\pi}{6}\right)+i\left(-1-\sin\frac{\pi}{6}\right)\right) =$$

$$= \frac{1}{6}\left(-\frac{\sqrt{3}}{2} - i\cdot\frac{3}{2}\right) = -\frac{\sqrt{3}}{12} - i\cdot\frac{3}{12}$$

$$\operatorname*{Res}_{z=z_1} f(z) = \frac{i^2+1}{6i} = 0$$

$$\operatorname*{Res}_{z=z_2} f(z) = \frac{e^{i\frac{5}{3}\pi}+1}{6e^{i\frac{25}{6}\pi}} = \frac{e^{i\frac{5}{3}\pi}}{6e^{i\frac{25}{6}\pi}} + \frac{1}{6e^{i\frac{25}{6}i}} = \frac{e^{-i\frac{5}{2}\pi}}{6} + \frac{1}{6e^{i\frac{25}{6}\pi}} =$$

$$\frac{1}{6}\left(e^{-i\frac{\pi}{2}} + e^{-i\frac{\pi}{6}}\right) = \frac{1}{6}\left(\cos\left(-\frac{\pi}{2}\right)+i\sin\left(-\frac{\pi}{2}\right)+\cos\left(-\frac{\pi}{6}\right)+i\sin\left(-\frac{\pi}{6}\right)\right)$$

$$= \frac{1}{6}\left(\frac{\sqrt{3}}{2} + i\left(-1-\frac{1}{2}\right)\right) = \frac{\sqrt{3}}{12} - i\cdot\frac{3}{12}$$

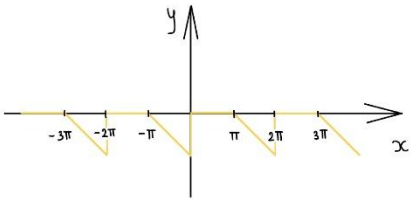
$$\int\limits_0^\infty \frac{x^2+1}{x^6+1}dx = \frac{1}{2}\int\limits_{+\infty}^{\infty} \frac{x^2+1}{x^6+1}dx = \frac{1}{2}\cdot 2\pi i\cdot\left(\cancel{-\frac{\sqrt{3}}{12}} - i\cdot\frac{3}{12} + \cancel{\frac{\sqrt{3}}{12}} - i\cdot\frac{3}{12}\right) =$$

$$\pi i\cdot -i\,\frac{2\cdot 3}{12} = \frac{\pi}{2}$$

4. Разложить функцию

$$f(x) = \begin{cases} 0, & x \in (0,\ \pi); \\ \pi - x, & x \in (\pi,\ 2\pi). \end{cases}$$

с периодом 2π в ряд Фурье. Построить графики функции $f(x)$ и первых частичных сумм $S_0(x)$, $S_1(x)$, $S_2(x)$, $S_3(x)$.



$$a_0 = \frac{1}{\pi}\int_0^{2\pi} f(x)\,dx =$$

$$\frac{1}{\pi}\int_0^{\pi} 0\,dx + \frac{1}{\pi}\int_{\pi}^{2\pi}(\pi - x)\,dx =$$

$$\frac{1}{\pi}\left(\pi x - \frac{x^2}{2}\right)\Bigg|_{\pi}^{2\pi} = \frac{1}{\pi}\left(2\pi^2 - \frac{4\pi^2}{2} - \left(\pi^2 - \frac{\pi^2}{2}\right)\right) = \frac{1}{\pi}\cdot\left(-\frac{\pi^2}{2}\right) = -\frac{\pi}{2}$$

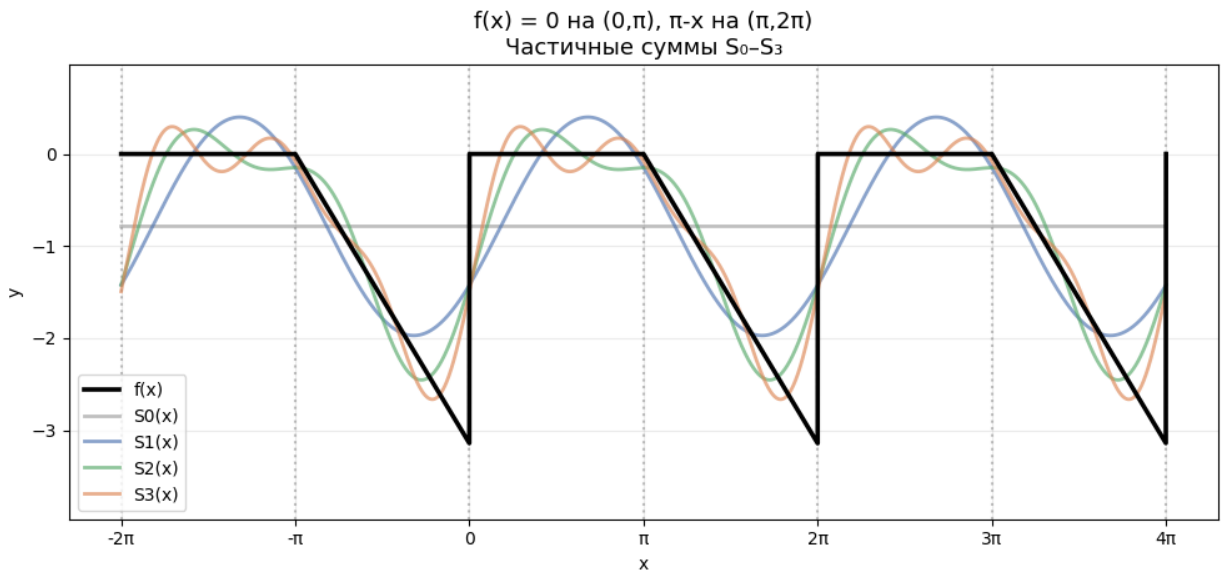
$$a_n = \frac{1}{\pi}\int_{\pi}^{2\pi}(\pi - x)\cos nx\,dx \quad \left[\begin{array}{ll} u = \pi - x, & du = -dx \\ dv = \cos nx\,dx, & v = \frac{\sin nx}{n} \end{array}\right] =$$

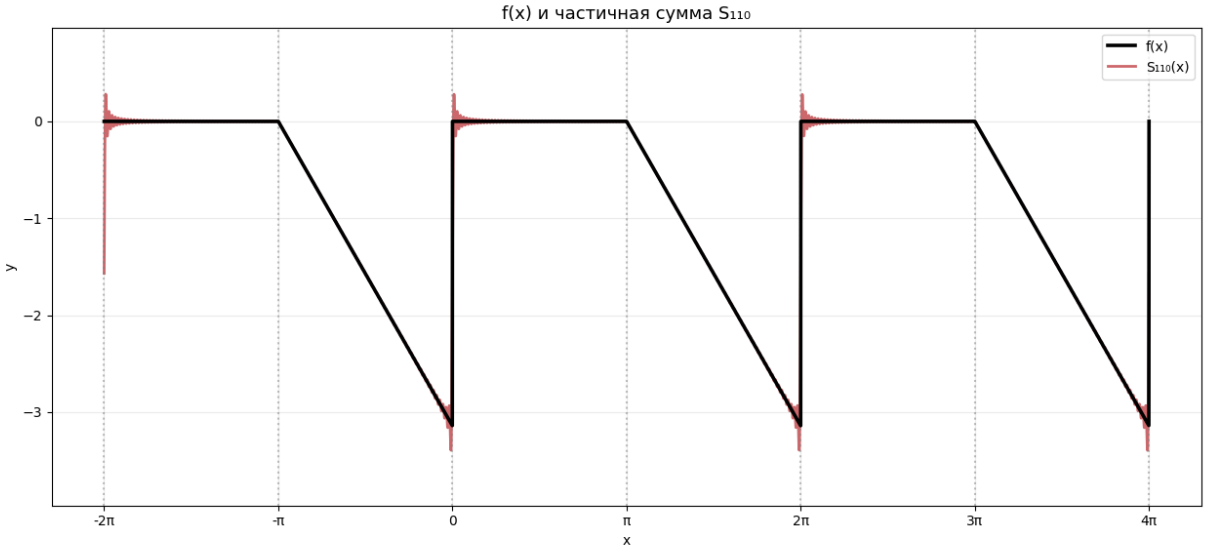
$$\frac{1}{\pi}\left((\pi - x)\cdot\frac{\sin nx}{n}\Bigg|_{\pi}^{2\pi} + \frac{1}{n}\int_{\pi}^{2\pi}\sin nx\,dx\right) = \frac{1}{\pi n}\cdot -\frac{\cos nx}{n}\Bigg|_{\pi}^{2\pi} = \frac{1}{\pi n^2}\left(-1 + (-1)^n\right)$$

$$b_n = \frac{1}{\pi}\int_{\pi}^{2\pi}(\pi - x)\sin nx\,dx \quad \left[\begin{array}{ll} u = \pi - x & du = -dx \\ dv = \sin nx\,dx & v = -\frac{\cos nx}{n} \end{array}\right] =$$

$$= \frac{1}{\pi}\left((\pi - x)\cdot -\frac{\cos nx}{n}\Bigg|_{\pi}^{2\pi} - \frac{1}{n}\int_{\pi}^{2\pi}\cos nx\,dx\right) = \frac{1}{\pi}\left(-\pi\cdot -\frac{1}{n} - \overbrace{\frac{1}{n}\cdot\frac{\sin nx}{n}\Bigg|_{\pi}^{2\pi}}^{=0}\right) = \frac{1}{n}$$

$$f(x) = -\frac{\pi}{4} + \sum_{n=1}^{\infty}\left(\frac{1}{\pi n^2}\left(-1 + (-1)^n\right)\cos\left(\frac{nxn}{\pi}\right) + \frac{1}{n}\cdot\sin\left(\frac{n\pi x}{\pi}\right)\right)$$

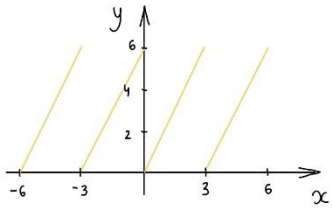




5. Разложить в ряд Фурье периодическую функцию

$$f(x) = 2x; \;\; x \in (0,\, 3)\,.$$

заданную на промежутке, длина которого равна периоду. Построить графики функции $f(x)$ и первых частичных сумм $S_0(x)$, $S_1(x)$, $S_2(x)$, $S_3(x)$.



$$a_0 = \frac{1}{1{,}5}\int_0^3 f(x)\,dx = \frac{2}{3}\int_0^3 2x\,dx = \frac{2}{3}\left(x^2\Big|_0^3\right) = \frac{2}{3}\cdot 9 = 6$$

$$a_n = \frac{1}{1{,}5}\int_0^3 2x\cdot\cos\left(\frac{n\pi x}{1{,}5\cdot 2}\right)dx = \frac{4}{3}\int_0^3 x\cos\left(\frac{n\pi x}{3}\right)dx \begin{bmatrix} u = x & du = dx \\ dv = \cos\left(\frac{n\pi x}{3}\right)dx & \\ v = \frac{3}{n\pi}\sin\left(\frac{n\pi x}{3}\right) & \end{bmatrix} =$$

$$\frac{4}{3}\left(\overbrace{\frac{3x}{n\pi}\sin\left(\frac{n\pi x}{3}\right)\Big|_0^3}^{=0} - \frac{3}{n\pi}\int_0^3 \sin\left(\frac{n\pi x}{3}\right)dx\right) = \frac{4}{3}\cdot\frac{3^2}{\pi^2 n^2}\cos\left(\frac{n\pi x}{3}\right)\Big|_0^3 =$$

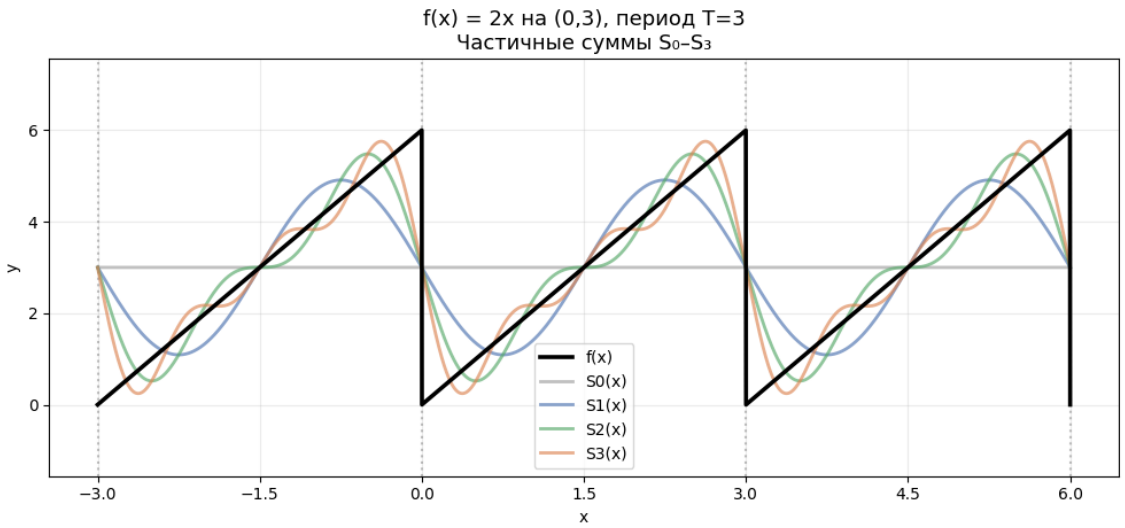
$$\frac{12}{\pi^2 n^2}\left(\cos(\pi n) - \cos(0)\right) = \frac{12}{\pi^2 n^2}\left((-1)^n - 1\right)$$

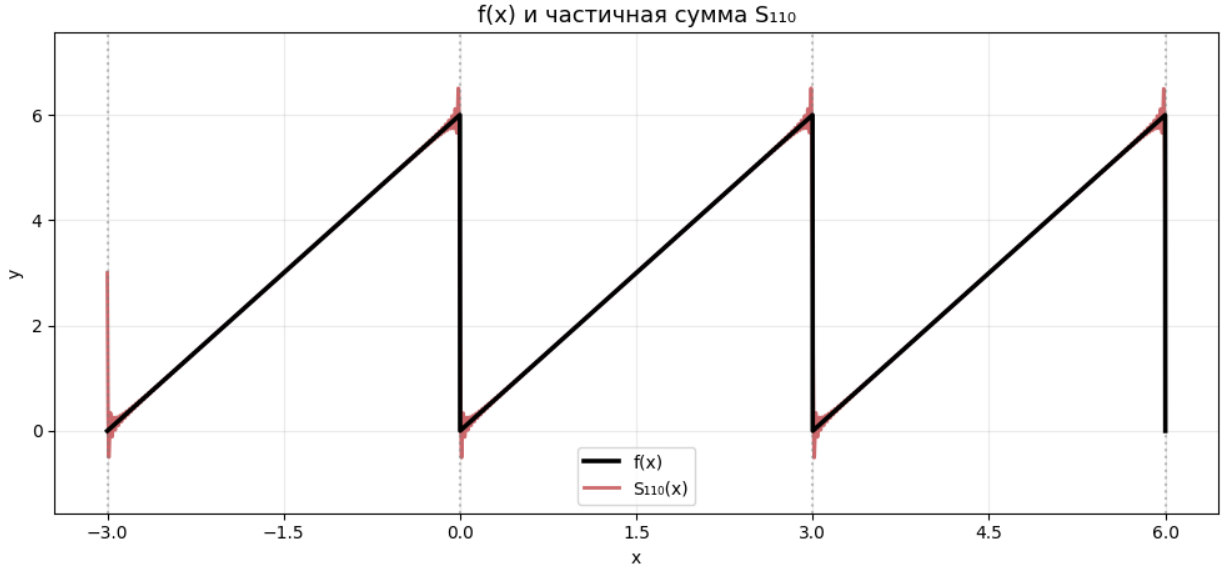
$$b_n = \frac{2}{3}\int_0^3 2x\cdot\sin\left(\frac{\pi n x}{3}\right)dx \begin{bmatrix} u = x & du = dx \\ dv = \sin\left(\frac{\pi n x}{3}\right) & \\ v = -\frac{3}{\pi n}\cos\left(\frac{\pi n x}{3}\right) & \end{bmatrix} = \frac{4}{3}\left(-\frac{3x}{\pi n}\cos\left(\frac{\pi n x}{3}\right)\Big|_0^3 - \int_0^3 -\frac{3}{\pi n}\cos\left(\frac{\pi n x}{3}\right)dx\right) =$$

$$= \frac{4}{3}\left(-\frac{9}{\pi n}\cos(\pi n) + \frac{3}{\pi n}\int_0^3 \cos\left(\frac{\pi n x}{3}\right)dx\right) = \frac{4}{3}\left(-\frac{9}{\pi n}(-1)^n + \frac{9}{\pi^2 n^2}\sin\left(\frac{\pi n x}{3}\right)\Big|_0^3\right) = \frac{4}{3}\left(-\frac{9}{\pi n}(-1)^n + \frac{9}{\pi^2 n^2}\cdot\sin(\pi n)\right) =$$

$$\frac{4}{\cancel{3}}\cdot -\frac{\overset{3}{\cancel{9}}}{\pi n}(-1)^n = -\frac{12}{\pi n}(-1)^n$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n\cos\left(\frac{\pi n x}{3}\right) + b_n\sin\left(\frac{\pi n x}{3}\right)\right) = 3 + \sum_{n=1}^{\infty}\frac{12}{\pi^2 n^2}\left((-1)^n - 1\right)\cos\left(\frac{\pi n x}{3}\right) - \frac{12}{\pi n}(-1)^n\sin\left(\frac{\pi n x}{3}\right)$$


f(x) = 2x на (0,3), период T=3
Частичные суммы S₀–S₃
f(x)
S0(x)
S1(x)
S2(x)
S3(x)
x
y


f(x) и частичная сумма S₁₁₀
f(x)
S₁₁₀(x)
x
y